

American International University-Bangladesh (AIUB)

Faculty of Science and Technology

Department of Computer Science

Web Technologies

CSE 3215

PROJECT REPORT

Project Title:

PharmaQuick – Online Medicine Shop

Semester: Spring 25-26

Submission Date: May 19, 2026

Section: H

Submitted To:

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Declaration: We hereby declare that this project report is our original work. Any external resources or references used have been properly cited. We have not engaged in any form of academic dishonesty.

Marks Sheet

To be filled by the instructor only. Students must NOT write in this section.

Group No.		Section	
Member 1 ID			

CO3: Illustrate Multi-Tier Web Application for Targeted Society					
Feature Implementation (0-3)			Feature Implementation (0-3)		Completeness (0-3)
Basic Web Security	UI (HTML/CSS)	Feature Completeness	DB	Auth (Session/cookie)	MVC
CO4: Coordinate with Group Members to Develop Multi-Tier Application					
Technical Knowledge (0-3)			Collaborative Teamwork (0-3)		Communication/Promptness (0-3)
JS validation	PHP Validation	Ajax/JSON	Git Contribution		VIVA
Expected Outcome (0-2)		CO3 Total	CO4 Total		Project Total

Evaluated By: _____ **Date:** _____

Course Teacher Signature: _____

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Chapter 1: Project Overview

1.1 Project Description

PharmaQuick is a multi-tier web application designed to simulate an online medicine marketplace. The platform addresses the hassle of visiting physical pharmacies by enabling customers to browse, search, and purchase medicines online from multiple vendors through a single, centralized system.

The application supports three primary user roles:

- Customer – Can register and log in, browse available medicines, manage a shopping cart, place orders, and review order history.
- Vendor (Pharmacy) – Can log in to manage medicine inventories, including adding, updating, and removing products, as well as viewing order records.
- Admin – Has full control over the platform, including managing medicines, categories, customers, vendors, and orders, along with monitoring the complete purchase history of all users.

Developed as a group academic project for the CSE 3215 – Web Technologies course at American International University-Bangladesh, the system demonstrates key full-stack development concepts such as MVC architecture, relational database management, session-based authentication, AJAX integration, and collaborative development using Git workflows.

1.2 Technologies Used

Layer	Technology / Tool	Purpose
Frontend	HTML5, CSS3	UI structure and responsive layout
	JavaScript (Vanilla)	Client-side form validation, AJAX for order status updates
Backend	PHP 8.2	Server-side logic, MVC pattern routing, session management
Database	MySQL	Persistent data storage for users, medicines, orders, payments
Auth	PHP Sessions & Cookies	Login state management; Admin gate checks
Version Control	Git / GitHub	Feature-branch workflow with Pull Requests
Server	XAMPP / Apache	Local development environment

1.3 System Architecture / MVC Diagram

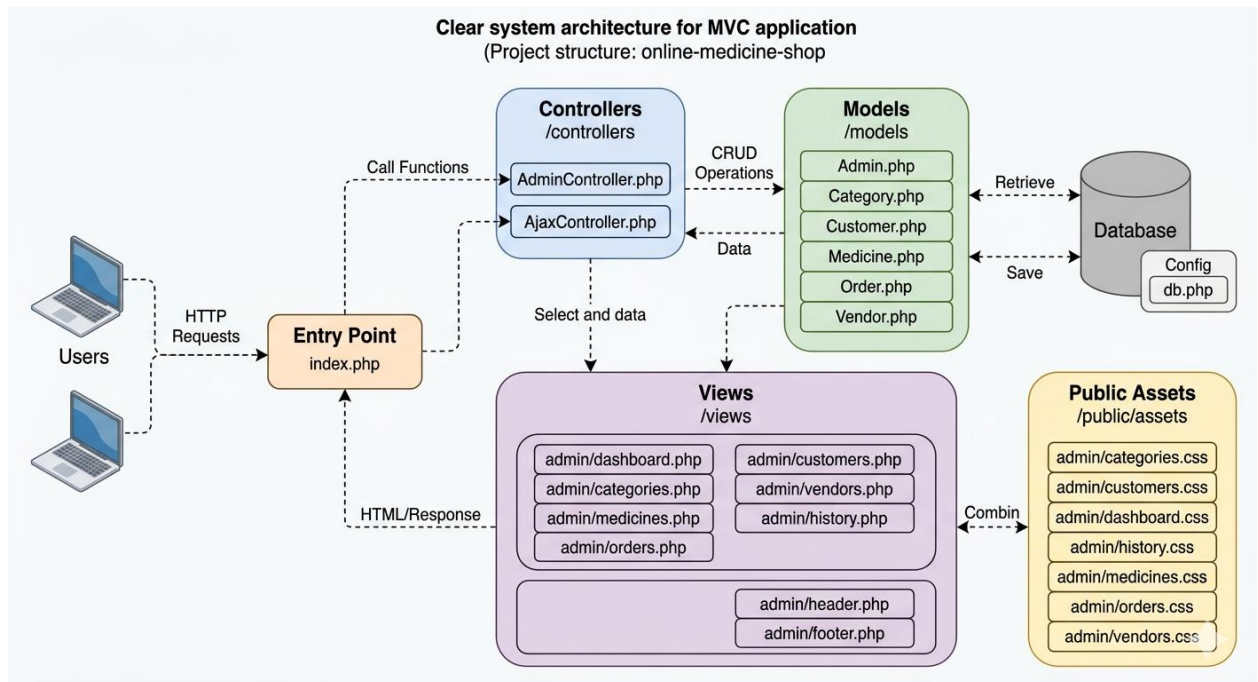


Fig: MVC Architecture Diagram

1.4 Database Schema

Table Name	Field Name	Data Type	Description
users	id	INT (PK)	Auto-increment user ID
	name	VARCHAR(100)	User full name
	email	VARCHAR(255)	Unique email
	password_hash	VARCHAR(255)	Bcrypt password hash
	role	ENUM	admin/customer/vendor
	address	TEXT	Shipping address
	phone	VARCHAR(20)	Contact number
categories	id	INT (PK)	Category ID
	name	VARCHAR(100)	Category name
	category_type	ENUM	solid/liquid
medicines	id	INT (PK)	Medicine ID
	name	VARCHAR(100)	Medicine name
	category_id	INT (FK)	References categories.id
	vendor_name	VARCHAR(100)	Vendor name
	price	DECIMAL(10,2)	Medicine price
	availability	INT	Stock quantity
	description	TEXT	Medicine description
	image	VARCHAR(255)	Medicine image path
orders	id	INT (PK)	Order ID
	customer_id	INT (FK)	References users.id
	total_amount	DECIMAL(10,2)	Total order amount
	status	ENUM	pending/accepted/rejected
	order_date	DATETIME	Order timestamp

Chapter 2: Git Contribution & Task Allocation

2.1 GitHub Repository Link

Repository URL: <https://github.com/nabil-adib/online-medicine-shop.git>

2.2 Git Contribution Screenshot

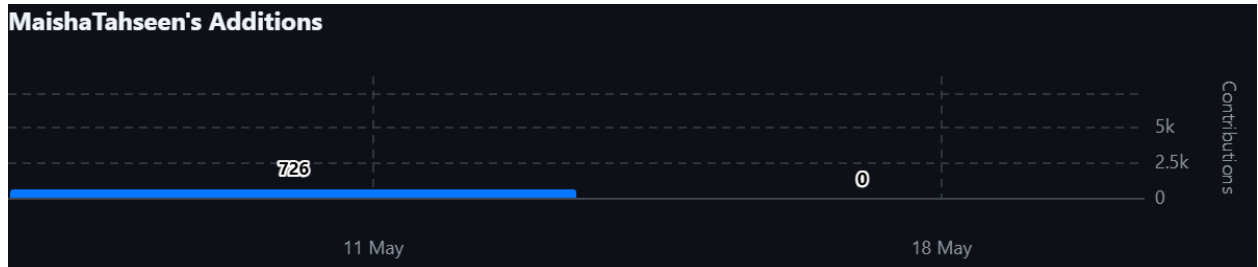


Fig: GitHub Contribution Graph

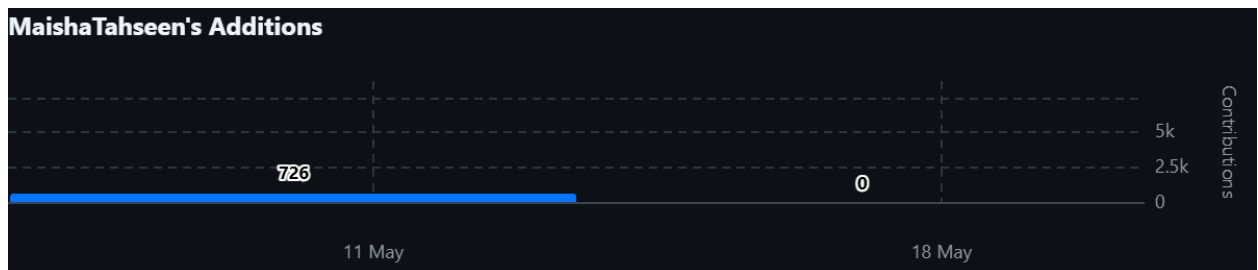


Fig: GitHub Commit History

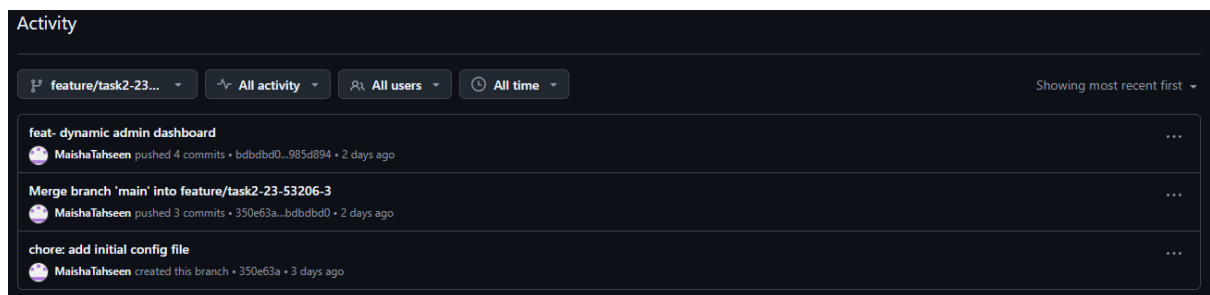


Fig: GitHub Commit Activity

Chapter 3: Features, Source Code & Screenshots

All features documented in this section were implemented as part of the assigned role of Platform Admin. The primary responsibilities include full CRUD operations on medicines and categories, managing users, and overseeing the order lifecycle.

Feature 1: Admin Dashboard

Description

Provides a high-level overview of the entire platform. It displays key performance indicators (total categories, medicines, customers, and pending orders) and a quick list of recent pending orders for immediate action.

Relevant Source Code

File: `Online_Medicine_Shop/controllers/AdminController.php`

```
if (!isset($_SESSION['user_role']) || $_SESSION['user_role'] !== 'admin') {
    header('Location: index.php?page=login');
    exit;
}

// Determine which action to perform based on URL parameter
$action = $_GET['action'] ?? 'dashboard';

// Route to appropriate handler based on action
switch ($action) {

    // DASHBOARD - Display admin home with statistics
    case 'dashboard':
        $categoriesCount = get_total_categories($conn);
        $medicinesCount   = get_total_medicines($conn);
        $customersCount   = get_total_customers($conn);
        $ordersCount      = count(get_pending_orders($conn));
        $recentOrders     = get_pending_orders($conn);
        require 'views/admin/dashboard.php';
        break;
```

Code Explanation

- A strict admin gate is implemented at the start of `admin_ctrl()`. It checks if `$_SESSION['user_role'] === 'admin'`. If not, the user is immediately redirected to the login page, securing all admin functions.
- The controller uses helper functions from models/ (like `get_total_medicines`) to fetch aggregated data from the database.
- The `$categoriesCount`, `$medicinesCount`, `$customersCount`, `$ordersCount` variables are passed to the `dashboard.php` view.
- The view renders these numbers inside stylized cards, providing an intuitive at-a-glance status of the system.

Screenshot(s)



[Dashboard](#) [Categories](#) [Medicines](#) [Customers](#) [Vendors](#) [Orders](#) [History](#) [Profile](#) [Logout](#)

Dashboard

CATEGORIES
7

MEDICINES
6

CUSTOMERS
2

PENDING ORDERS
2

Pending Orders

ORDER ID	CUSTOMER	STATUS	DATE
#14	Nabil Adib	pending	2026-05-18
#15	Nabil Adibur	pending	2026-05-18

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Feature 2: Category Management (Full CRUD)

Description

Allows the admin to create, read, update, and delete medicine categories. Each category has a name and a category_type (solid or liquid) for logical segmentation. The system **blocks deletion** if any medicine is still assigned to that category, preventing data integrity issues.

Relevant Source Code

File: Online_Medicine_Shop/controllers/AdminController.php

```
// Add Category
case 'add_category':
    if ($_SERVER['REQUEST_METHOD'] === 'POST') {
        // Validation...
        add_category($conn, $name, $type);
        header('Location: ...&msg=added');
    }
    break;

// Delete Category with Check
case 'delete_category':
    if (check_medicines_in_category($conn, $id)) {
        header('Location: ...&msg=blocked');
        exit;
    }
    delete_category($conn, $id);
    header('Location: ...&msg=deleted');
    break;
```

File: Online_Medicine_Shop/models/Category.php

```
function check_medicines_in_category($conn, $category_id) {
    $stmt = mysqli_prepare($conn, "SELECT COUNT(*) AS total FROM medicines WHERE
category_id = ?");
    mysqli_stmt_bind_param($stmt, "i", $category_id);
    // ...
    return $row['total'] > 0; // Returns true if medicines exist
}
```

Code Explanation

- The `add_category` action validates the input (e.g., name not empty) before inserting into the categories table.
- The `edit_category` action fetches existing category data and pre-fills the form for updating.
- The `delete_category` action first calls `check_medicines_in_category()`. This function runs a `SELECT COUNT(*)` query to see if any medicine in the medicines table references the category ID.
- If the count is greater than 0, the deletion is aborted and the user is redirected with an error message (`msg=blocked`). This ensures a clean relational database state.
- Success or error messages are displayed in the view using `$_GET['msg']` parameters.

Screenshot(s)



[Dashboard](#) [Categories](#) [Medicines](#) [Customers](#) [Vendors](#) [Orders](#) [History](#) [Profile](#) [Logout](#)

Add New Category

CATEGORY NAME

Enter category name

CATEGORY TYPE

Select Type

Add Category

All Categories

ID	NAME	CATEGORY TYPE	ACTIONS
1	Antibiotics	Solid	Edit Delete
2	Analgesics & Pain Relief	Solid	Edit Delete
3	Antihistamines	Liquid	Edit Delete
4	Cough Syrups	Liquid	Edit Delete

Feature 3: Medicine Management (Full CRUD)

Description

Enables the admin to add, view, edit, and delete medicine listings. The medicine creation form includes fields for name, category (dropdown), vendor name, price (>0), availability (stock), description, and an optional **image upload** (JPEG/PNG ≤2MB). Editing pre-fills the form, and deletion checks if the medicine is part of any pending order before removal.

Relevant Source Code

File: Online_Medicine_Shop/controllers/AdminController.php

```
case 'add_medicine':
    // Validation...
    $imagePath = '';
    if (!empty($_FILES['image_path']['name'])) {
        $imagePath = 'public/uploads/' . time() . '_' .
$_FILES['image_path']['name'];
        move_uploaded_file($_FILES['image_path']['tmp_name'], $imagePath);
    }
    add_medicine($conn, $data);
    break;
```

File: Online_Medicine_Shop/controllers/AdminController.php

```
function can_delete_medicine($conn, $medicine_id) {
    // Check if in any pending order
    $order_check = mysqli_query($conn, "SELECT COUNT(*) FROM order_items oi
        JOIN orders o ON oi.order_id = o.id
        WHERE oi.medicine_id = $medicine_id AND
o.status = 'pending'");
    $order_count = mysqli_fetch_row($order_check)[0];
    return $order_count == 0;
}
```

Code Explanation

- The form in `views/admin/medicines.php` is dynamically reused for both **Add** and **Edit** (\$isEdit flag) to reduce code duplication.
- For image upload, a unique filename is generated using `time()` to prevent overwriting. The file is moved to the `public/uploads/` directory, and its path is stored in the database.
- The `can_delete_medicine()` function performs a critical check. It joins the `order_items` and `orders` tables to see if the medicine is present in any order with a pending status.
- If the medicine is in a pending order, the admin is shown an error message, and the delete action is halted. This prevents customer orders from referencing a deleted product.
- Server-side validation ensures price and availability are numeric values greater than zero.

Screenshot(s)

The screenshot displays the PharmaQuick web application's 'Medicine Management' section. The top navigation bar includes the PharmaQuick logo and links to Dashboard, Categories, Medicines, Customers, Vendors, Orders, History, Profile, and Logout. The main content area is titled 'Medicine Management' and features a form for adding new medicines. The form includes fields for Medicine Name, Category (a dropdown menu), Vendor Name, Price, Stock, and Image (with a 'Choose File' button and 'No file chosen' text). A Description field is also present. An 'Add Medicine' button is located at the bottom of the form. Below the form, there is a section titled 'All Medicines' which is currently empty.

PharmaQuick

Dashboard Categories Medicines Customers Vendors Orders History Profile Logout

Medicine Management

Add Medicine

Medicine Name

Category

Vendor Name

Price

Stock

Image

Description

Add Medicine

All Medicines

Feature 4: Customer and Vendor Management

Description

Allows the admin to view a list of all registered customers and vendors. The admin can delete customer accounts. The deletion process is cascade safe, meaning it automatically removes the customer's cart items and all associated orders before deleting the user record. The admin can also add, edit and delete vendors.

Relevant Source Code

File: `Online_Medicine_Shop/models/Customer.php`

```
function delete_customer($conn, $id) {
    // 1. delete cart items
    mysqli_query($conn, "DELETE FROM cart WHERE user_id = $id");
    // 2. get all orders of this user
    $orders = mysqli_query($conn, "SELECT id FROM orders WHERE user_id = $id");
    while ($order = mysqli_fetch_assoc($orders)) {
        // 3. delete order items for each order
        mysqli_query($conn, "DELETE FROM order_items WHERE order_id = {$order['id']}");
        // 4. delete the order
        mysqli_query($conn, "DELETE FROM orders WHERE id = {$order['id']}");
    }
    // 5. finally, delete the user
    return mysqli_query($conn, "DELETE FROM users WHERE id = $id AND role = 'customer'");
}
```

File: `Online_Medicine_Shop/models/Vendor.php`

```
function add_vendor($conn, $name, $email, $password_hash, $address, $phone)
{
    $role = 'vendor';
    $stmt = mysqli_prepare($conn, "INSERT INTO users (name, email, password_hash, address, phone, role) VALUES (?, ?, ?, ?, ?, ?)");
    mysqli_stmt_bind_param($stmt, 'ssssss', $name, $email, $password_hash, $address, $phone, $role);
    $result = mysqli_stmt_execute($stmt);
    mysqli_stmt_close($stmt);
    return $result;
}
```

Code Explanation

- Sequential deletion process follows a strict order from child to parent records, preventing foreign key constraint violations in the relational database structure
- Cascade safety guarantee ensures that a customer cannot be deleted while leaving behind orphaned orders or cart items, maintaining complete database consistency.
- Role hardcoding automatically sets the user role to 'vendor' within the function, preventing unauthorized role assignment from client-side manipulation.

Screenshot(s)



DashboardCategoriesMedicinesCustomersVendorsOrdersHistoryProfileLogout

Customer Management

All Customers

ID	NAME	EMAIL	ACTIONS
10	Nabil Adibur	nabiladib@gmail.com	Delete
7	Nabil Adib	nabil@gmail.com	Delete

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DashboardCategoriesMedicinesCustomersVendorsOrdersHistoryProfileLogout

FULL NAME

Enter vendor name

PHONE NUMBER

01712345678

EMAIL ADDRESS

vendor@example.com

PASSWORD

Min 8 characters

CONFIRM PASSWORD

Repeat password

ADDRESS

Vendor's full address

Add Vendor

All Vendors

1 Total

ID	NAME	EMAIL	PHONE	ADDRESS	JOINED DATE	ACTIONS
9	Nabil Adib	nabilvendor111@gmail.com	01987654321	banani 112	May 18, 2026	EditDelete

Feature 5: Order Management (Accept/Reject via AJAX)

Description

Displays a comprehensive list of all purchase requests. For each pending order, the admin can **Accept** or **Reject** it. This status update is performed **asynchronously using AJAX**, providing a seamless user experience without a full page reload.

Relevant Source Code

File: `Online_Medicine_Shop/views/admin/orders.php`

```
function updateStatus(orderId, status) {
    fetch('index.php?page=admin&action=update_order_status&order_id=' + orderId +
    '&status=' + status, {
        credentials: 'same-origin'
    })
    .then(response => response.json())
    .then(data => {
        if (data.success) {
            location.reload(); // Reload to update the status badge
        } else {
            alert('Update failed');
        }
    });
}
```

File: `Online_Medicine_Shop/controllers/AdminController.php`

```
case 'update_order_status':
    header('Content-Type: application/json');
    $order_id = $_GET['order_id'] ?? 0;
    $status = $_GET['status'] ?? '';
    if ($order_id && in_array($status, ['accepted', 'rejected'])) {
        $result = update_order_status($conn, $order_id, $status);
        echo json_encode(['success' => $result]);
    } else {
        echo json_encode(['success' => false]);
    }
    exit;
```

Code Explanation

- When an admin clicks "Accept" or "Reject", the `updateStatus()` JavaScript function is triggered.
- This function sends an asynchronous GET request (using the `fetch` API) to the same `index.php`, but with `page=admin&action=update_order_status` and the relevant `order_id` and `status`.
- The `AdminController` captures this action, validates the status (only 'accepted' or 'rejected'), and calls `update_order_status()` to update the orders table.
- The controller returns a JSON response (`{ "success": true/false }`), which the JavaScript receives.
- On a successful response, `location.reload()` is called to refresh the page and visually update the order's status badge.

Screenshot(s)



[Dashboard](#) [Categories](#) [Medicines](#) [Customers](#) [Vendors](#) [Orders](#) [History](#) [Profile](#) [Logout](#)

Order Management

All Purchase Requests

ID	CUSTOMER	SHIPPING ADDRESS	TOTAL	DATE	STATUS	ACTION
11	Nabil Adib	Mirpur 1216	195.00	2026-05-18	Accepted	—
12	Nabil Adib	Mirpur 1216	360.00	2026-05-18	Rejected	—
13	Nabil Adib	Mirpur 1216	360.00	2026-05-18	Accepted	—
14	Nabil Adib	Mirpur 1216	120.00	2026-05-18	Pending	Accept Reject
15	Nabil Adibur	Gulshan	240.00	2026-05-18	Pending	Accept Reject

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Feature 6: Purchase History Viewer

Description

A dedicated admin page (history.php) that displays a detailed report of all completed (accepted) orders. It shows the customer's information (name, email, phone, address) and a breakdown of each item purchased (medicine name, quantity, unit price, subtotal), along with the order total..

Relevant Source Code

File: `Online_Medicine_Shop/views/admin/history.php`

```
<?php foreach ($orders as $order): ?>
    <!-- Individual Order Container -->
    <div class="order-box">

        <!-- Order Header Section with ID and Date -->
        <div class="order-header">
            <h3>Order #<?= $order['order_id'] ?></h3>
            <p class="order-date">Date: <?= date('d M Y, h:i A',
strtotime($order['order_date'])) ?></p>
        </div>

        <!-- Customer Information Section -->
        <div class="customer-info">
            <h4>Customer Information</h4>
            <p><strong>Name:</strong> <?=
htmlspecialchars($order['customer_name']) ?></p>
            <p><strong>Email:</strong> <?=
htmlspecialchars($order['customer_email']) ?></p>
            <p><strong>Phone:</strong> <?=
htmlspecialchars($order['customer_phone']) ?? 'N/A' ?></p>
            <p><strong>Address:</strong> <?=
htmlspecialchars($order['shipping_address']) ?></p>
            <p><strong>Payment Method:</strong> <?= ucfirst(str_replace('_',
' ', $order['payment_method'])) ?></p>
        </div>

        <!-- Order Items Table Section -->
        <h4>Medicines Ordered</h4>
        <table class="order-table">
            <thead>
                <tr>
                    <th>Medicine Name</th>
                    <th>Quantity</th>
                    <th>Unit Price</th>
                    <th>Subtotal</th>
                </tr>
            </thead>
            <tbody>
                <!-- Loop through each medicine in the order -->
                <?php foreach ($order['items'] as $item): ?>
                    <tr>
                        <!-- Medicine name with HTML escaping -->
                        <td><?= htmlspecialchars($item['medicine_name'])
?></td>

                        <!-- Quantity ordered -->
                        <td><?= $item['quantity'] ?></td>
                        <!-- Unit price formatted with 2 decimals -->
                        <td><?= number_format($item['unit_price'], 2)
?></td>

                        <!-- Subtotal calculation result -->
                        <td><?= number_format($item['subtotal'], 2) ?></td>
                    </tr>
                <?php endforeach; ?>
            </tbody>
        </table>
    </div>
?>
```

```

        <!-- Grand Total Row -->
        <tr class="total-row">
            <td colspan="3"><strong>Grand Total</strong></td>
            <td><strong>৳ <?= number_format($order['total_amount'],
2) ?></strong></td>
        </tr>
    </tbody>
</table>
</div>

    <!-- Separator between orders -->
    <hr class="order-separator">
    <?php endforeach; ?>

```

Code Explanation

- The function `get_completed_orders_with_details()` uses an **INNER JOIN** across four tables (orders, users, order_items, medicines) to fetch all necessary data in a single, efficient query.
- The result set is then reorganized in PHP into a nested associative array, where each order object contains a sub-array of its items. This makes the data easy to loop through in the view.
- The `history.php` view iterates through this data, generating an ordered, human-readable report for the admin.
- This feature provides full financial and logistical transparency, which is crucial for an admin.

Screenshot(s)

The screenshot displays the PharmaQuick admin dashboard. The top navigation bar includes links for Dashboard, Categories, Medicines, Customers, Vendors, Orders, History, Profile, and Logout. The main content area shows two summary cards: 'COMPLETED ORDERS' with a value of 2, and 'TOTAL REVENUE' with a value of ৳ 480.00. Below these, the 'Order #13' details are shown, including the date '18 May 2026, 12:00 AM'. The 'Customer Information' section lists: Name: Nabil Adib, Email: nabil@gmail.com, Phone: 017625364781, Address: Mirpur 1216, and Payment Method: Cash on Delivery. The 'Medicines Ordered' section contains a table with the following data:

MEDICINE NAME	QUANTITY	UNIT PRICE	SUBTOTAL
Fenadin 120mg	4	৳ 90.00	৳ 360.00
Grand Total			৳ 360.00

Chapter 4: Conclusion & Future Work

4.1 Summary

For this project, I successfully implemented the complete Platform Admin Module for the PharmaQuick Online Medicine Shop. My responsibilities included developing the core administrative functionalities required to manage the e-commerce ecosystem.

The specific features implemented are:

- Admin Dashboard: A centralized view of system-wide statistics and recent pending orders.
- Category and Medicine Management (Full CRUD): Allowing admins to create, read, update, and delete product categories and medicines, with a safety mechanism to block deletion of categories that contain medicines.
- Customer and Vendor Management: Providing admins with the ability to view and delete user accounts, with a cascade-safe deletion process for customers.
- Order Management: A comprehensive interface for viewing all orders and updating their status (Accept/Reject) using asynchronous AJAX requests.
- Purchase History Viewer: A detailed reporting page for completed orders, giving admins full visibility into all financial transactions.

The primary goals achieved were to build a secure, efficient, and user-friendly back-office interface that provides complete control over the platform's data and operations.

4.2 Challenges Faced

- Secure File Uploads: Validating uploaded images to ensure they were legitimate image files (using MIME type) and not malicious scripts, while also managing unique file names to prevent overwrites.
- Cascading Deletion Logic: Implementing a safe and reliable `delete_customer()` function that removes a user and all their relational data (cart, orders, order items) without causing foreign key constraint errors in MySQL.
- Conditional Blocking on Delete: Developing the logic to block the deletion of a medicine if it's part of a pending order, or blocking the deletion of a category if it has any medicines assigned. This required multi-table checks before performing the delete operation.
- AJAX State Management: Ensuring the `update_order_status()` function correctly handled the asynchronous request, validated the data, interacted with the database, and returned a proper JSON response without redirecting or breaking the session.

4.3 Future Improvements

- Advanced Reporting and Analytics: Implement graphical dashboards using libraries like Chart.js to visualize sales trends, top-selling medicines, and revenue over different time periods (daily, weekly, monthly).
- Bulk Operations: Add the ability for admins to perform bulk actions, such as deleting multiple medicines, changing categories in bulk, or exporting order data to CSV/Excel files.
- Admin Notification System: Implement a real-time notification system (e.g., using WebSockets or Server-Sent Events) to instantly alert the admin when a new pending order is placed by a customer.
- Role-Specific Permissions: Further refine the admin module by introducing sub-roles (e.g., 'Inventory Manager', 'Order Processor', 'Finance Admin'), where each has granular permissions over different sections of the admin panel.

References

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- [4] OWASP Foundation. (2024). *SQL Injection Prevention Cheat Sheet*. https://cheatsheetseries.owasp.org/cheatsheets/SQL_Injection_Prevention_Cheat_Sheet.html